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NUMERICAL METHODS

CPE 312

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Lecturer



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BAIRSTOW'S METHOD



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Learning Objectives

- *Define Bairstow's Method**
- *Determine and perform the algorithm of Bairstow's Method**
- *Solve functions using Bairstow's Method**



- An iterative numerical technique used in numerical analysis to find the roots of a polynomial equation.
- An extension of the Newton-Raphson method but adapted to handle quadratic factors.



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- It was formulated by Leonard Bairstow, a British engineer and mathematician.
- He developed this numerical method in the early 20th century as a practical tool to find roots of polynomials, especially in engineering applications where real and complex roots needed efficient computation.



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- It was particularly useful when the polynomial has real coefficients but may have complex roots.



Bairstow's Method Algorithm

1. Initialize r and s
2. Find a , b and c
3. Compute corrections Δr and Δs .
4. Update r and s

$$r = r + \Delta r$$

$$s = s + \Delta s$$



5. Calculate the approximate error for r and s until it will close to zero.

6. Solve the roots using

$$x = \frac{r \pm \sqrt{r^2 + 4s}}{2}$$



Example:

Find the roots of polynomial

$$f(x) = 9x^4 + 12x^3 + 13x^2 + 12x + 4 \text{ and}$$

$$r = -1, s = -1.$$



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Solution:

$$f(x) = 9x^4 + 12x^3 + 13x^2 + 12x + 4$$

Initial approximation: $r = -1$, $s = -1$



Solution:

$$f(x) = 9x^4 + 12x^3 + 13x^2 + 12x + 4$$

Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$



Iteration 1:

Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = 9$$



Iteration 1:

Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = 9$$

$$b_3 = a_3 + rb_4$$



Iteration 1:

Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = 9$$

$$b_3 = a_3 + rb_4 = 12 + (-1)(9) = 3$$



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Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = \mathbf{9}$$

$$b_3 = a_3 + rb_4 = 12 + (-1)(9) = \mathbf{3}$$

$$b_2 = a_2 + rb_3 + sb_4$$



Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = \mathbf{9}$$

$$b_3 = a_3 + rb_4 = 12 + (-1)(9) = \mathbf{3}$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-1)(3) + (-1)(9) = \mathbf{1}$$



Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_3 = a_3 + rb_4 = 12 + (-1)(9) = \mathbf{3}$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-1)(3) + (-1)(9) = \mathbf{1}$$

$$b_1 = a_1 + rb_2 + sb_3$$



Initial approximation: $r = -1, s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_3 = a_3 + rb_4 = 12 + (-1)(9) = \mathbf{3}$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-1)(3) + (-1)(9) = \mathbf{1}$$

$$b_1 = a_1 + rb_2 + sb_3 = 12 + (-1)(1) + (-1)(3) = \mathbf{8}$$



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Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-1)(3) + (-1)(9) = \mathbf{1}$$

$$b_1 = a_1 + rb_2 + sb_3 = 12 + (-1)(1) + (-1)(3) = \mathbf{8}$$

$$b_0 = a_0 + rb_1 + sb_2$$



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Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-1)(3) + (-1)(9) = \mathbf{1}$$

$$b_1 = a_1 + rb_2 + sb_3 = 12 + (-1)(1) + (-1)(3) = \mathbf{8}$$

$$b_0 = a_0 + rb_1 + sb_2 = 4 + (-1)(8) + (-1)(1) = \mathbf{-5}$$



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Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9$$

$$a_3 = 12$$

$$a_2 = 13$$

$$a_1 = 12$$

$$a_0 = 4$$

$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = b_4 = 9$$



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Initial approximation: $r = -1$, $s = -1$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 3$	$b_2 = 1$	$b_1 = 8$	$b_0 = -5$

$$c_4 = b_4 = 9$$

$$c_3 = b_3 + rc_4$$



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Initial approximation: $r = -1$, $s = -1$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 3$	$b_2 = 1$	$b_1 = 8$	$b_0 = -5$

$$c_4 = b_4 = 9$$

$$c_3 = b_3 + rc_4 = 3 + (-1)(9) = -6$$



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Initial approximation: $r = -1$, $s = -1$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 3$	$b_2 = 1$	$b_1 = 8$	$b_0 = -5$

$$c_4 = b_4 = \mathbf{9}$$

$$c_3 = b_3 + rc_4 = 3 + (-1)(9) = \mathbf{-6}$$

$$c_2 = b_2 + rc_3 + sc_4$$



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Initial approximation: $r = -1$, $s = -1$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 3$	$b_2 = 1$	$b_1 = 8$	$b_0 = -5$

$$c_4 = b_4 = \mathbf{9}$$

$$c_3 = b_3 + rc_4 = 3 + (-1)(9) = \mathbf{-6}$$

$$c_2 = b_2 + rc_3 + sc_4 = 1 + (-1)(-6) + (-1)(9) = \mathbf{-2}$$



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Initial approximation: $r = -1$, $s = -1$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 3$	$b_2 = 1$	$b_1 = 8$	$b_0 = -5$

$$c_4 = b_4 = \mathbf{9}$$

$$c_3 = b_3 + rc_4 = 3 + (-1)(9) = \mathbf{-6}$$

$$c_2 = b_2 + rc_3 + sc_4 = 1 + (-1)(-6) + (-1)(9) = \mathbf{-2}$$

$$c_1 = b_1 + rc_2 + sc_3$$



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Initial approximation: $r = -1$, $s = -1$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 3$	$b_2 = 1$	$b_1 = 8$	$b_0 = -5$

$$c_4 = b_4 = \mathbf{9}$$

$$c_3 = b_3 + rc_4 = 3 + (-1)(9) = \mathbf{-6}$$

$$c_2 = b_2 + rc_3 + sc_4 = 1 + (-1)(-6) + (-1)(9) = \mathbf{-2}$$

$$c_1 = b_1 + rc_2 + sc_3 = 8 + (-1)(-2) + (-1)(-6) = \mathbf{16}$$



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Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9$$

$$a_3 = 12$$

$$a_2 = 13$$

$$a_1 = 12$$

$$a_0 = 4$$

$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

Cramer's rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$



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Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9$$

$$a_3 = 12$$

$$a_2 = 13$$

$$a_1 = 12$$

$$a_0 = 4$$

$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

Cramer's rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (-2)(-2) - (16)(-6) = 100$$



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Initial approximation: $r = -1$, $s = -1$

$$a_4 = 9$$

$$a_3 = 12$$

$$a_2 = 13$$

$$a_1 = 12$$

$$a_0 = 4$$

$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

Cramer's rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (-2)(-2) - (16)(-6) = 100$$

$$D_1 = (b_0 \times c_3) - (b_1 \times c_2)$$



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$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

Cramer's rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (-2)(-2) - (16)(-6) = 100$$

$$D_1 = (b_0 \times c_3) - (b_1 \times c_2)$$

$$D_1 = (-5)(-6) - (8)(-2) = 46$$



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$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

Cramer's rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (-2)(-2) - (16)(-6) = 100$$

$$D_1 = (b_0 \times c_3) - (b_1 \times c_2)$$

$$D_1 = (-5)(-6) - (8)(-2) = 46$$

$$D_2 = (b_1 \times c_1) - (b_0 \times c_2)$$



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$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

Cramer's rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (-2)(-2) - (16)(-6) = 100$$

$$D_1 = (b_0 \times c_3) - (b_1 \times c_2)$$

$$D_1 = (-5)(-6) - (8)(-2) = 46$$

$$D_2 = (b_1 \times c_1) - (b_0 \times c_2)$$

$$D_2 = (8)(16) - (-5)(-2) = 118$$



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$$a_4 = 9$$

$$a_3 = 12$$

$$a_2 = 13$$

$$a_1 = 12$$

$$a_0 = 4$$

$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

$$D = 100$$

$$D_1 = 46$$

$$D_2 = 118$$

$$\Delta r = \frac{D_1}{D}$$



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$$a_4 = 9$$

$$a_3 = 12$$

$$a_2 = 13$$

$$a_1 = 12$$

$$a_0 = 4$$

$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

$$D = 100$$

$$D_1 = 46$$

$$D_2 = 18$$

$$\Delta r = \frac{D_1}{D} = \frac{46}{100} = 0.46$$



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$$a_4 = 9$$

$$a_3 = 12$$

$$a_2 = 13$$

$$a_1 = 12$$

$$a_0 = 4$$

$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

$$D = 100$$

$$D_1 = 46$$

$$D_2 = 118$$

$$\Delta r = \frac{D_1}{D} = \frac{46}{100} = 0.46$$

$$\Delta s = \frac{D_2}{D}$$



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$$a_4 = 9$$

$$a_3 = 12$$

$$a_2 = 13$$

$$a_1 = 12$$

$$a_0 = 4$$

$$b_4 = 9$$

$$b_3 = 3$$

$$b_2 = 1$$

$$b_1 = 8$$

$$b_0 = -5$$

$$c_4 = 9$$

$$c_3 = -6$$

$$c_2 = -2$$

$$c_1 = 16$$

$$D = 100$$

$$D_1 = 46$$

$$D_2 = 118$$

$$\Delta r = \frac{D_1}{D} = \frac{46}{100} = 0.46$$

$$\Delta s = \frac{D_2}{D} = \frac{118}{100} = 1.18$$



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Solve for new r and s :

$$r = r + \Delta r$$



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Solve for new r and s :

$$r = r + \Delta r$$

$$r = -1 + 0.46 = -0.54$$



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Solve for new r and s :

$$r = r + \Delta r$$

$$r = -1 + 0.46 = -0.54$$

$$s = s + \Delta s$$



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Solve for new r and s:

$$r = r + \Delta r$$

$$r = -1 + 0.46 = -0.54$$

$$s = s + \Delta s$$

$$s = -1 + 1.18 = 0.18$$



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Approximate error for r and s:

$$|\varepsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$



Approximate error for r and s:

$$|\varepsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$
$$|\varepsilon_{a,r}| = \left| \frac{0.46}{-0.54} \right| \times 100\%$$



Approximate error for r and s:

$$|\varepsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$
$$|\varepsilon_{a,r}| = \left| \frac{0.46}{-0.54} \right| \times 100\%$$
$$|\varepsilon_{a,r}| = 85.18519\%$$



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Approximate error for r and s:

$$|\epsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\% \qquad |\epsilon_{a,s}| = \left| \frac{\Delta s}{s} \right| \times 100\%$$

$$|\epsilon_{a,r}| = \left| \frac{0.46}{-0.54} \right| \times 100\%$$

$$|\epsilon_{a,r}| = 85.18519\%$$



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Approximate error for r and s:

$$|\epsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$

$$|\epsilon_{a,r}| = \left| \frac{0.46}{-0.54} \right| \times 100\%$$

$$|\epsilon_{a,r}| = 85.18519\%$$

$$|\epsilon_{a,s}| = \left| \frac{\Delta s}{s} \right| \times 100\%$$

$$|\epsilon_{a,s}| = \left| \frac{1.18}{0.18} \right| \times 100\%$$



Approximate error for r and s:

$$|\epsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$

$$|\epsilon_{a,r}| = \left| \frac{0.46}{-0.54} \right| \times 100\%$$

$$|\epsilon_{a,r}| = 85.18519\%$$

$$|\epsilon_{a,s}| = \left| \frac{\Delta s}{s} \right| \times 100\%$$

$$|\epsilon_{a,s}| = \left| \frac{1.18}{0.18} \right| \times 100\%$$

$$|\epsilon_{a,s}| = 655.55556\%$$



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Iteration 2: $r = -0.54$, $s = 0.18$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = \mathbf{9}$$

$$b_3 = a_3 + rb_4$$



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Iteration 2: $r = -0.54$, $s = 0.18$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = \mathbf{9}$$

$$b_3 = a_3 + rb_4 = 12 + (-0.54)(9) = 7.14$$



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Iteration 2: $r = -0.54$, $s = 0.18$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = \mathbf{9}$$

$$b_3 = a_3 + rb_4 = 12 + (-0.54)(9) = 7.14$$

$$b_2 = a_2 + rb_3 + sb_4$$



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Iteration 2: $r = -0.54, s = 0.18$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = \mathbf{9}$$

$$b_3 = a_3 + rb_4 = 12 + (-0.54)(9) = 7.14$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-0.54)(7.14) + (0.18)(9) = 10.7644$$



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Iteration 2: $r = -0.54, s = 0.18$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = \mathbf{9}$$

$$b_3 = a_3 + rb_4 = 12 + (-0.54)(9) = 7.14$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-0.54)(7.14) + (0.18)(9) = 10.7644$$

$$b_1 = a_1 + rb_2 + sb_3$$



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Iteration 2: $r = -0.54$, $s = 0.18$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = 9$$

$$b_3 = a_3 + rb_4 = 12 + (-0.54)(9) = 7.14$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-0.54)(7.14) + (0.18)(9) = 10.7644$$

$$b_1 = a_1 + rb_2 + sb_3 = 12 + (-0.54)(10.7644) + (0.18)(7.14) = 7.47242$$



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Iteration 2: $r = -0.54$, $s = 0.18$

$$a_4 = 9 \quad a_3 = 12 \quad a_2 = 13 \quad a_1 = 12 \quad a_0 = 4$$

$$b_4 = a_4 = 9$$

$$b_3 = a_3 + rb_4 = 12 + (-0.54)(9) = 7.14$$

$$b_2 = a_2 + rb_3 + sb_4 = 13 + (-0.54)(7.14) + (0.18)(9) = 10.7644$$

$$b_1 = a_1 + rb_2 + sb_3 = 12 + (-0.54)(10.7644) + (0.18)(7.14) = 7.47242$$

$$b_0 = a_0 + rb_1 + sb_2$$



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Iteration 2: $r = -0.54, s = 0.18$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 7.14$	$b_2 = 10.7644$	$b_1 = 7.47242$	$b_0 = 1.90249$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \end{array}$$

$$c_4 = b_4 = 9$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \end{array}$$

$$c_4 = b_4 = 9$$

$$c_3 = b_3 + rc_4$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \end{array}$$

$$c_4 = b_4 = 9$$

$$c_3 = b_3 + rc_4 = 7.14 + (-0.54)(9) = 2.28$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \end{array}$$

$$c_4 = b_4 = 9$$

$$c_3 = b_3 + rc_4 = 7.14 + (-0.54)(9) = 2.28$$

$$c_2 = b_2 + rc_3 + sc_4$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \end{array}$$

$$c_4 = b_4 = 9$$

$$c_3 = b_3 + rc_4 = 7.14 + (-0.54)(9) = 2.28$$

$$c_2 = b_2 + rc_3 + sc_4 = 10.7644 + (-0.54)(2.28) + (0.18)(9) = 11.1532$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \end{array}$$

$$c_4 = b_4 = 9$$

$$c_3 = b_3 + rc_4 = 7.14 + (-0.54)(9) = 2.28$$

$$c_2 = b_2 + rc_3 + sc_4 = 10.7644 + (-0.54)(2.28) + (0.18)(9) = 11.1532$$

$$c_1 = b_1 + rc_2 + sc_3$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \end{array}$$

$$c_4 = b_4 = \mathbf{9}$$

$$c_3 = b_3 + rc_4 = 7.14 + (-0.54)(9) = 2.28$$

$$c_2 = b_2 + rc_3 + sc_4 = 10.7644 + (-0.54)(2.28) + (0.18)(9) = 11.1532$$

$$c_1 = b_1 + rc_2 + sc_3 = 7.47242 + (-0.54)(11.1532) + (0.18)(2.28) = 1.86009$$



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Iteration 2: $r = -0.54, s = 0.18$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 7.14$	$b_2 = 10.7644$	$b_1 = 7.47242$	$b_0 = 1.90249$
$c_4 = 9$	$c_3 = 2.28$	$c_2 = 11.1532$	$c_1 = 1.86009$	



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Iteration 2: $r = -0.54, s = 0.18$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 7.14$	$b_2 = 10.7644$	$b_1 = 7.47242$	$b_0 = 1.90249$
$c_4 = 9$	$c_3 = 2.28$	$c_2 = 11.1532$	$c_1 = 1.86009$	

$$c_2 \Delta r + c_3 \Delta s = -b_1$$

$$c_1 \Delta r + c_2 \Delta s = -b_0$$



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Iteration 2: $r = -0.54, s = 0.18$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 7.14$	$b_2 = 10.7644$	$b_1 = 7.47242$	$b_0 = 1.90249$
$c_4 = 9$	$c_3 = 2.28$	$c_2 = 11.1532$	$c_1 = 1.86009$	

$$c_2 \Delta r + c_3 \Delta s = -b_1$$

$$c_1 \Delta r + c_2 \Delta s = -b_0$$

$$11.1532 \Delta r + 2.28 \Delta s = -7.47242$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$c_2 \Delta r + c_3 \Delta s = -b_1$$

$$11.1532 \Delta r + 2.28 \Delta s = -7.47242$$

$$c_1 \Delta r + c_2 \Delta s = -b_0$$

$$1.86009 \Delta r + 11.1532 \Delta s = -1.90249$$



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Iteration 2: $r = -0.54, s = 0.18$

$a_4 = 9$	$a_3 = 12$	$a_2 = 13$	$a_1 = 12$	$a_0 = 4$
$b_4 = 9$	$b_3 = 7.14$	$b_2 = 10.7644$	$b_1 = 7.47242$	$b_0 = 1.90249$
$c_4 = 9$	$c_3 = 2.28$	$c_2 = 11.1532$	$c_1 = 1.86009$	

Cramer's Rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{lllll} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

Cramer's Rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (11.1532)(11.1532) - (1.86009)(2.28) = 120.15287$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

Cramer's Rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (11.1532)(11.1532) - (1.86009)(2.28) = 120.15287$$

$$D_1 = (b_0 \times c_3) - (b_1 \times c_2)$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

Cramer's Rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (11.1532)(11.1532) - (1.86009)(2.28) = 120.15287$$

$$D_1 = (b_0 \times c_3) - (b_1 \times c_2)$$

$$D_1 = (1.90249)(2.28) - (7.47242)(11.1532) = -79.00372$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{lllll} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

Cramer's Rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (11.1532)(11.1532) - (1.86009)(2.28) = 120.15287$$

$$D_1 = (b_0 \times c_3) - (b_1 \times c_2)$$

$$D_1 = (1.90249)(2.28) - (7.47242)(11.1532) = -79.00372$$

$$D_2 = (b_1 \times c_1) - (b_0 \times c_2)$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{lllll} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

Cramer's Rule:

$$D = (c_2 \times c_2) - (c_1 \times c_3)$$

$$D = (11.1532)(11.1532) - (1.86009)(2.28) = 120.15287$$

$$D_1 = (b_0 \times c_3) - (b_1 \times c_2)$$

$$D_1 = (1.90249)(2.28) - (7.47242)(11.1532) = -79.00372$$

$$D_2 = (b_1 \times c_1) - (b_0 \times c_2)$$

$$D_2 = (7.47242)(1.86009) - (1.90249)(11.1532) = -7.31948$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287 \quad D_1 = -79.00372 \quad D_2 = -7.31948$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287$$

$$D_1 = -79.00372$$

$$D_2 = -7.31948$$

$$\Delta r = \frac{D_1}{D}$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287 \quad D_1 = -79.00372 \quad D_2 = -7.31948$$

$$\Delta r = \frac{D_1}{D} = \frac{-79.00372}{120.15287} = -0.65753$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287 \quad D_1 = -79.00372 \quad D_2 = -7.31948$$

$$\Delta r = \frac{D_1}{D} = \frac{-79.00372}{120.15287} = -0.65753$$

$$\Delta s = \frac{D_2}{D}$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287 \quad D_1 = -79.00372 \quad D_2 = -7.31948$$

$$\Delta r = \frac{D_1}{D} = \frac{-79.00372}{120.15287} = -0.65753$$

$$\Delta s = \frac{D_2}{D} = \frac{-7.31948}{120.15287} = -0.06092$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287 \quad D_1 = -79.00372 \quad D_2 = -7.31948$$

$$\Delta r = \frac{D_1}{D} = \frac{-79.00372}{120.15287} = -0.65753 \quad r = r + \Delta r$$

$$\Delta s = \frac{D_2}{D} = \frac{-7.31948}{120.15287} = -0.06092$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287 \quad D_1 = -79.00372 \quad D_2 = -7.31948$$

$$\Delta r = \frac{D_1}{D} = \frac{-79.00372}{120.15287} = -0.65753 \quad r = r + \Delta r$$

$$\Delta s = \frac{D_2}{D} = \frac{-7.31948}{120.15287} = -0.06092 \quad r = -0.54 + (-0.65753)$$

$$r = -1.19753$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287 \quad D_1 = -79.00372 \quad D_2 = -7.31948$$

$$\Delta r = \frac{D_1}{D} = \frac{-79.00372}{120.15287} = -0.65753 \quad \mathbf{s = s + \Delta s}$$

$$\Delta s = \frac{D_2}{D} = \frac{-7.31948}{120.15287} = -0.06092$$



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Iteration 2: $r = -0.54, s = 0.18$

$$\begin{array}{ccccc} a_4 = 9 & a_3 = 12 & a_2 = 13 & a_1 = 12 & a_0 = 4 \\ b_4 = 9 & b_3 = 7.14 & b_2 = 10.7644 & b_1 = 7.47242 & b_0 = 1.90249 \\ c_4 = 9 & c_3 = 2.28 & c_2 = 11.1532 & c_1 = 1.86009 & \end{array}$$

$$D = 120.15287 \quad D_1 = -79.00372 \quad D_2 = -7.31948$$

$$\Delta r = \frac{D_1}{D} = \frac{-79.00372}{120.15287} = -0.65753 \quad \mathbf{s = s + \Delta s}$$

$$\Delta s = \frac{D_2}{D} = \frac{-7.31948}{120.15287} = -0.06092 \quad \mathbf{s = 0.18 + (-0.06092)}$$

$$\mathbf{s = 0.11908}$$



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$$r = -1.19753 \qquad s = 0.11908$$

Approximate error for r and s:

$$|\varepsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$



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$$r = -1.19753 \quad s = 0.11908$$

Approximate error for r and s:

$$|\varepsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$

$$|\varepsilon_{a,r}| = \left| \frac{-0.65753}{-1.19753} \right| \times 100\%$$



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$$r = -1.19753 \qquad s = 0.11908$$

Approximate error for r and s:

$$|\varepsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$

$$|\varepsilon_{a,r}| = \left| \frac{-0.65753}{-1.19753} \right| \times 100\%$$

$$|\varepsilon_{a,r}| = 54.90718 \%$$



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$$r = -1.19753 \qquad s = 0.11908$$

Approximate error for r and s:

$$|\varepsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\% \qquad |\varepsilon_{a,s}| = \left| \frac{\Delta s}{s} \right| \times 100\%$$

$$|\varepsilon_{a,r}| = \left| \frac{-0.65753}{-1.19753} \right| \times 100\%$$

$$|\varepsilon_{a,r}| = 54.90718 \%$$



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$$r = -1.19753 \quad s = 0.11908$$

Approximate error for r and s:

$$|\varepsilon_{a,r}| = \left| \frac{\Delta r}{r} \right| \times 100\%$$

$$|\varepsilon_{a,r}| = \left| \frac{-0.65753}{-1.19753} \right| \times 100\%$$

$$|\varepsilon_{a,r}| = 54.90718 \%$$

$$|\varepsilon_{a,s}| = \left| \frac{\Delta s}{s} \right| \times 100\%$$

$$|\varepsilon_{a,s}| = \left| \frac{-0.06092}{0.11908} \right| \times 100\%$$

$$|\varepsilon_{a,s}| = 51.15888 \%$$



ACTIVITY # 10:

Continue the iterations of the given function until the approximate error is equal to 0% then solve for the approximate root using

$$x = \frac{r \pm \sqrt{r^2 + 4s}}{2} .$$



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Pointers to Review

Introduction to Numerical Methods

Graphical Method

Incremental-Search Method

Bisection Method

False-Position Method

Newton-Raphson Method

Secant Method

Muller's Method

Bairstow's Method



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